

AAP-001 — Why AI Needs Action Paths

Sizhe Tan

with chatGPT (OpenAI)

2026-07-06

Repository: <https://github.com/sizhet/AI-Action-Paths-AAP>

Abstract

The remarkable progress of Large Language Models (LLMs) has transformed artificial intelligence from a research topic into a widely deployed technological infrastructure. Much of the current AI landscape, however, remains centered on a single question:

How capable can an AI model become?

This question has driven tremendous advances in model scaling, reasoning benchmarks, coding performance, and multimodal understanding.

AAP argues that an equally important question is now emerging:

What sustainable actions can AI reliably perform in the real world?

This repository proposes that the next stage of AI evolution should be understood not only through model capability, but also through **Action Paths**—persistent, executable, trustworthy, and socially meaningful forms of AI participation.

From Capability to Action

Current AI evaluation is largely capability-oriented.

Typical metrics include:

- reasoning benchmarks
- coding benchmarks
- mathematical competitions
- scientific examinations
- multimodal understanding

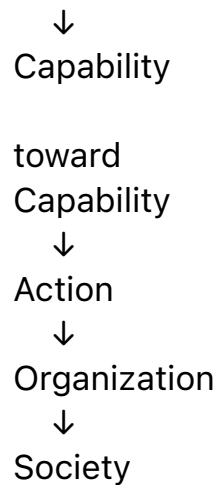
These measurements are valuable because they estimate the potential capability of a model.

However, capability alone does not determine how AI participates in society.

Two systems with similar benchmark scores may contribute very differently in practice depending on the actions they can continuously perform.

AAP therefore shifts the primary perspective from

Model



The objective is not to replace capability evaluation, but to complement it with an action-oriented perspective.

AI Evolution Through Action Paths

Every significant stage of AI development has expanded the space of executable actions.

For example,

- search engines expanded information retrieval,
- recommendation systems expanded personalized information delivery,
- LLMs expanded natural language interaction.

These are not merely improvements in model architecture.

They are expansions of AI action space.

AAP studies this expansion systematically.

An **Action Path** is defined as a stable and reusable form of AI participation that can repeatedly create value under a trusted runtime.

Examples include:

- LLM scaling
- Human–AI discussion
- Brain Units
- AI freelance workers
- autonomous research systems
- hybrid human–AI organizations

Each Action Path represents a different way in which intelligence becomes productive.

Why Action Paths Matter

As AI systems become increasingly capable, the limiting factor gradually shifts.

The question is no longer simply

Can AI solve this problem?

but increasingly

Can AI continuously perform this role?

This distinction is fundamental.

Many impressive demonstrations remain isolated achievements.

Action Paths instead focus on continuous participation, accumulated experience, trusted execution, and organizational integration.

They therefore bridge the gap between isolated intelligence and sustainable productivity.

AAP Is a General Framework

AAP is not proposed as an alternative to Large Language Models.

Instead, it provides a higher-level organizational framework covering multiple AI paradigms.

It includes:

- Large Language Models
- Structural Intelligence
- Runtime Intelligence
- Brain Units
- Hybrid AI systems
- Future autonomous architectures

Different AI technologies may implement different Action Paths, while sharing common structural foundations.

Structural Foundations

Throughout this repository, Action Paths are viewed as being supported by reusable structural assets rather than isolated model behaviors.

Examples include:

- Runtime Invariants (RI)
- Calling Graphs
- Differential Trees
- Common Concept Cores (CCC)
- Function Tunnels (FT)
- Universal Task Naming (UTN)

These structural components enable actions to become repeatable, verifiable, and transferable.

AAP therefore studies not only what AI can do, but also how sustainable AI actions are constructed.

Repository Goal

This repository explores AI from the perspective of executable action rather than model capability alone.

Its objective is to provide an organizational roadmap connecting

- AI models,
- trusted runtimes,
- Brain Units,
- autonomous workers,
- collective learning,
- and future Human–AI civilizations.

Rather than predicting a single future architecture, AAP investigates how increasingly rich Action Paths may gradually reshape the evolution of artificial intelligence.

Key Takeaways

- AI capability and AI action are complementary concepts.
- Future AI development depends on expanding sustainable Action Paths.
- Action Paths connect intelligence with organization and society.
- Existing AI technologies already represent different Action Paths.
- Structural assets make trusted and reusable Action Paths possible.
- AAP provides an organizational perspective for understanding future AI evolution.